

LP-V PRACTICAL ORAL EXAM — FULL MARKS MASTER GUIDE

High Performance Computing (HPC) + Deep Learning (DL)

Prepared from:

- LP-V Viva Question Bank
 - LP-V Lab Manual
 - HPC & Deep Learning Oral Exam Preparation Guide
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HOW TO SCORE FULL MARKS IN ORAL

External examiner mainly checks:

- Confidence
- Clarity of concepts
- Ability to explain code
- Basic practical understanding
- Communication skills

BEST WAY TO ANSWER

Always answer in this order:

1. Definition
2. Working
3. Important concept
4. Parallelism/Model explanation
5. Applications

Example:

“BFS stands for Breadth First Search. It is a graph traversal algorithm that visits nodes level by level using queue. In our experiment we used OpenMP for parallelism. BFS is mainly used in shortest path problems and web crawling.”

This answer structure sounds professional and easy to understand.

IMPORTANT ABBREVIATIONS

Short Form	Full Form
BFS	Breadth First Search
DFS	Depth First Search
CNN	Convolutional Neural Network
RNN	Recurrent Neural Network
DNN	Deep Neural Network
CUDA	Compute Unified Device Architecture
GPU	Graphics Processing Unit
CPU	Central Processing Unit
OpenMP	Open Multi-Processing
MSE	Mean Squared Error
NLP	Natural Language Processing
LSTM	Long Short-Term Memory

PART A — HIGH PERFORMANCE COMPUTING (HPC)

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EXPERIMENT 1(A) — PARALLEL BFS USING OPENMP

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Q1. What is BFS?

Answer: BFS stands for Breadth First Search. It is a graph traversal algorithm that visits nodes level by level. It first visits all neighboring nodes before moving to the next level.

BFS uses Queue data structure.

Applications:

- Shortest path finding
- GPS navigation

- Web crawling
 - Social network analysis
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Q2. Why Queue is used in BFS?

Answer: Queue follows FIFO (First In First Out) order.

In BFS, nodes are visited level by level. The first inserted node should be processed first, so queue is used.

Q3. Difference between BFS and DFS?

BFS	DFS
Level-wise traversal	Depth-wise traversal
Uses Queue	Uses Stack
Finds shortest path	Does not guarantee shortest path
Better for shortest path	Better for deep traversal

Q4. How is Parallel BFS implemented using OpenMP?

Answer: In Parallel BFS, nodes at the same level are processed simultaneously using multiple threads.

We use:

```
#pragma omp parallel for
```

This divides work among multiple CPU threads.

Q5. What is OpenMP?

Answer: OpenMP is an API used for parallel programming in C/C++.

It uses compiler directives like:

```
#pragma omp parallel
#pragma omp for
```

OpenMP helps programs run faster using multicore CPUs.

Q6. What is race condition?

Answer: Race condition occurs when multiple threads access shared data simultaneously and produce incorrect results.

Q7. Why do we use critical section?

Answer: Critical section prevents multiple threads from accessing shared resources at the same time.

```
#pragma omp critical
```

It avoids race condition.

Q8. Advantages of Parallel BFS

Answer:

- Faster execution
- Better CPU utilization
- Efficient for large graphs
- Reduces processing time

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EXPERIMENT 1(B) — PARALLEL DFS USING OPENMP

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Q1. What is DFS?

Answer: DFS stands for Depth First Search.

It explores nodes deeply before backtracking.

DFS uses Stack data structure.

Applications:

- Cycle detection
 - Maze solving
 - Topological sorting
 - Connected components
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Q2. Why Stack is used in DFS?

Answer: Stack follows LIFO (Last In First Out) order.

DFS explores one branch deeply and then returns back, so stack is suitable.

Q3. What is backtracking?

Answer: Backtracking means returning to previous node when no unvisited adjacent node exists.

Q4. How is Parallel DFS implemented?

Answer: In Parallel DFS, adjacent nodes are explored simultaneously using OpenMP threads.

Multiple threads process different branches together.

Q5. Why parallelizing DFS is difficult?

Answer: Parallelizing DFS is difficult because:

- DFS is recursive
 - Synchronization is required
 - Load balancing becomes difficult
 - Parent-child dependency exists
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Q6. Which is better for shortest path — BFS or DFS?

Answer: BFS is better because it explores nodes level by level and guarantees shortest path.

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EXPERIMENT 2 — PARALLEL SORTING

PARALLEL BUBBLE SORT

Q1. What is Bubble Sort?

Answer: Bubble Sort repeatedly compares adjacent elements and swaps them if they are in wrong order.

Q2. Time Complexity of Bubble Sort?

Answer: $O(n^2)$

Bubble Sort becomes slow for large datasets.

Q3. Why Bubble Sort is inefficient?

Answer: Because it performs many comparisons and swaps.

Q4. How is Parallel Bubble Sort implemented?

Answer: Different pairs of elements are compared simultaneously using OpenMP threads.

Odd-even sorting approach is commonly used.

Q5. Difference between Sequential and Parallel Bubble Sort?

Sequential	Parallel
Single thread	Multiple threads
Slower	Faster
No synchronization	Synchronization required

Q6. Applications of Bubble Sort

Answer:

- Educational purpose
 - Small datasets
 - Learning sorting algorithms
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PARALLEL MERGE SORT

Q1. What is Merge Sort?

Answer: Merge Sort divides array into smaller parts, sorts them separately, and merges them.

It uses Divide and Conquer technique.

Q2. Time Complexity of Merge Sort?

Answer: $O(n \log n)$

It is much faster than Bubble Sort.

Q3. Why Merge Sort better than Bubble Sort?

Answer: Merge Sort has lower time complexity and performs better for large datasets.

Q4. How is Merge Sort parallelized?

Answer: Different subarrays are sorted simultaneously using multiple threads.

Q5. Performance metrics used in parallel algorithms?

Answer:

- Execution time
- Speedup

- Efficiency
 - Scalability
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Q6. What is Speedup?

Answer: Speedup measures improvement in performance.

Formula:

Speedup = Sequential Time / Parallel Time

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EXPERIMENT 3 — PARALLEL REDUCTION

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Q1. What is Parallel Reduction?

Answer: Parallel Reduction combines multiple values into a single value.

Examples:

- Sum
 - Average
 - Minimum
 - Maximum
-

Q2. Why Parallel Reduction important?

Answer: It reduces execution time for large datasets.

Q3. Explain Sum Reduction.

Answer: 1. Array is divided among threads 2. Each thread computes partial sum 3. Partial sums are combined 4. Final sum is produced

Q4. What is reduction clause in OpenMP?

Answer: Reduction clause safely combines thread results.

```
#pragma omp parallel for reduction(+:sum)
```

Q5. Difference between Reduction and Critical Section?

Reduction	Critical
Faster	Slower
Optimized	Sequential access
Automatic combine	Manual synchronization

EXPERIMENT 4 — CUDA PROGRAMMING

Q1. What is CUDA?

Answer: CUDA is NVIDIA's GPU programming platform used for parallel computing.

It allows GPU to perform large number of operations simultaneously.

Q2. Difference between CUDA and OpenMP?

CUDA	OpenMP
GPU based	CPU based
Thousands of threads	Multithreading
Massive parallelism	Shared-memory parallelism

Q3. Why GPU faster than CPU?

Answer: GPU contains thousands of smaller cores that perform parallel operations simultaneously.

CPU contains fewer powerful cores.

Q4. What are Threads, Blocks, and Grids?

Term	Meaning
Thread	Smallest execution unit
Block	Group of threads
Grid	Collection of blocks

Q5. What is CUDA Kernel?

Answer: Kernel is a function executed on GPU.

Kernel launch syntax:

```
kernel<<<blocks, threads>>>();
```

Q6. Explain Vector Addition in CUDA.

Answer: Each GPU thread computes one output element.

Example:

```
C[i] = A[i] + B[i]
```

Q7. Explain Matrix Multiplication in CUDA.

Answer: Each GPU thread computes one element of output matrix.

Rows and columns are processed simultaneously.

Q8. Explain CUDA Memory Hierarchy.

Memory Type	Description
Global Memory	Large but slow
Shared Memory	Fast memory inside block
Registers	Fastest memory
Local Memory	Private memory per thread

Q9. Why Shared Memory important?

Answer: Shared memory reduces slow global memory access and improves performance.

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IMPORTANT GENERAL HPC QUESTIONS

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Q1. What is synchronization?

Answer: Synchronization coordinates multiple threads safely.

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Q2. Difference between CPU and GPU?

CPU	GPU
Few powerful cores	Thousands of smaller cores
Sequential tasks	Parallel tasks

Q3. What is scalability?

Answer: Scalability means improving performance using more processors.

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PART B — DEEP LEARNING

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EXPERIMENT 1 — LINEAR REGRESSION USING DNN

Q1. What is Linear Regression?

Answer: Linear Regression predicts continuous numerical values.

Example:

- House price prediction
 - Temperature prediction
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Q2. Formula of Linear Regression

$$y = mx + b$$

Where:

- y = output
 - m = slope
 - x = input
 - b = bias/intercept
-

Q3. What is Boston Housing Dataset?

Answer: Boston Housing Dataset contains 506 house samples with 13 features used for house price prediction.

Q4. What is MSE Loss Function?

Answer: MSE means Mean Squared Error.

It measures difference between predicted and actual values.

Formula:

$$\text{MSE} = (1/n) \sum (y_{\text{actual}} - y_{\text{predicted}})^2$$

Q5. Why MSE used?

Answer: MSE penalizes larger errors more heavily and helps model learn better.

Q6. Steps in training DNN

Answer: 1. Data preprocessing 2. Build model 3. Compile model 4. Train model 5. Validate model 6. Test predictions

Q7. How to prevent overfitting?

Answer:

- Dropout
- Regularization
- Early stopping
- More training data

EXPERIMENT 2 — CLASSIFICATION USING DNN

Q1. Difference between Regression and Classification?

Regression	Classification
Predicts continuous values	Predicts categories
Uses MSE	Uses Cross-Entropy
Example: price prediction	Example: spam detection

Q2. Difference between Binary and Multiclass Classification?

Binary	Multiclass
Two classes	More than two classes
Example: Yes/No	Example: Digit recognition

Q3. Activation function for Binary Classification?

Answer: Sigmoid.

Q4. Activation function for Multiclass Classification?

Answer: Softmax.

Q5. What is One-Hot Encoding?

Answer: One-Hot Encoding converts categorical labels into binary vectors.

Q6. What is Cross-Entropy Loss?

Answer: Cross-Entropy measures difference between predicted probability and actual class.

Q7. Explain IMDB Dataset.

Answer: IMDB Dataset contains movie reviews used for sentiment analysis.

Goal: Classify reviews as positive or negative.

Q8. Explain OCR Dataset.

Answer: OCR dataset contains handwritten letters used for multiclass classification.

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EXPERIMENT 3 — CNN

Q1. What is CNN?

Answer: CNN stands for Convolutional Neural Network.

It is mainly used for image processing and computer vision.

Applications:

- Face recognition
- Plant disease detection
- Medical imaging
- Self-driving cars

Q2. Why CNN suitable for images?

Answer: CNN automatically extracts spatial features like edges, textures, and patterns.

Q3. Explain Convolution Layer.

Answer: Filters slide over image and extract important features.

Q4. What are Filters/Kernels?

Answer: Filters are small matrices used to detect image features.

Q5. What is Pooling?

Answer: Pooling reduces image size and computation.

Q6. Difference between Max Pooling and Average Pooling?

Max Pooling	Average Pooling
Takes maximum value	Takes average value

Q7. Difference between CNN and Dense Neural Network?

CNN	Dense Network
Good for images	Good for tabular data
Shared weights	Fully connected
Fewer parameters	More parameters

Q8. Explain MNIST Fashion Dataset.

Answer: MNIST Fashion dataset contains clothing images of 10 categories.

Q9. Explain Plant Disease Detection using CNN.

Answer: CNN learns leaf patterns and classifies healthy and diseased plants.

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EXPERIMENT 4 — RNN / LSTM

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Q1. What is RNN?

Answer: RNN stands for Recurrent Neural Network.

It is designed for sequential or time-series data.

Applications:

- Stock prediction
 - Speech recognition
 - NLP
 - Language translation
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Q2. Difference between RNN and Standard Neural Network?

RNN	Standard Neural Network
Has memory	No memory
Handles sequence data	Handles fixed input
Learns temporal patterns	No temporal learning

Q3. What is Hidden State?

Answer: Hidden state stores information from previous time steps.

It acts like memory.

Q4. What is Vanishing Gradient Problem?

Answer: Gradients become very small during backpropagation through many time steps.

This makes learning difficult.

Q5. What is LSTM?

Answer: LSTM is advanced RNN architecture that remembers long-term information.

Q6. Gates in LSTM

Answer:

- Input Gate
 - Forget Gate
 - Output Gate
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Q7. Why LSTM better than RNN?

Answer: LSTM can learn long-term dependencies better and solves vanishing gradient problem.

Q8. Explain Time Series Prediction using RNN.

Answer: Historical sequence data is used to predict future values.

Example: Using previous stock prices to predict next stock price.

Q9. Explain Google Stock Dataset.

Answer: Google Stock Dataset contains historical stock prices used for stock prediction.

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GENERAL DEEP LEARNING QUESTIONS

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Q1. What is Backpropagation?

Answer: Backpropagation updates neural network weights using gradients.

Q2. What is Dropout?

Answer: Dropout randomly disables neurons during training to reduce overfitting.

Q3. Difference between Epoch, Batch Size, and Iteration?

Term	Meaning
Epoch	One complete pass of dataset
Batch Size	Samples processed together
Iteration	One batch update

Q4. Common Activation Functions?

Function	Use
ReLU	Hidden layers

Function	Use
Sigmoid	Binary classification
Softmax	Multiclass classification

MINI PROJECT QUESTIONS

Q1. Explain your project briefly.

Prepare:

- Problem statement
- Dataset used
- Model used
- Accuracy achieved
- Output

Q2. Why did you choose this project?

Answer: Because it solves a real-world problem and improved practical understanding.

Q3. Challenges faced during project?

Answer:

- Data preprocessing
- Overfitting
- Hardware limitations
- Training time

Q4. Future improvements?

Answer:

- Larger dataset
- Better accuracy

- Real-time deployment
- Better optimization

MOST IMPORTANT HIGH-PROBABILITY QUESTIONS

HPC

- OpenMP
- CUDA
- Reduction
- BFS vs DFS
- Threads/Blocks/Grids
- Race condition
- Matrix multiplication

Deep Learning

- CNN
- RNN vs CNN
- Overfitting
- Backpropagation
- Activation functions
- Loss functions
- Optimizers

FINAL ORAL EXAM TIPS

1. Understand concepts instead of memorizing only.
2. Revise code logic and outputs.
3. Know datasets used in each experiment.
4. Be able to compare algorithms/models.
5. Speak slowly and confidently.
6. Never stay silent in viva.
7. If you don't know exact answer, explain related concept.
8. Maintain eye contact and confidence.

BIGGEST SECRET TO SCORE FULL MARKS

Even if your answer is not perfect:

- speak confidently
- explain clearly
- use technical keywords
- show practical understanding

Most externals give marks based on:

- confidence
- clarity
- communication
- basic technical understanding

NOT based on perfect textbook answers.

ALL THE BEST FOR YOUR PRACTICAL EXAM
